How Long Documents Paginate

A whitepaper on line-level pagination: when a paragraph crosses a page boundary, the engine moves whole lines to the next page so no glyph is ever split.

1. Line breaking

Line breaking happens at the whitespace between words. Each line is measured against the available content width and grows until the next word no longer fits; that word starts the following line. This means a paragraph is a sequence of lines, and the pagination machinery only ever cuts between lines, never through them.

The engine measures every run with the embedded Liberation Sans font at the resolved font size, applying letter-spacing when declared. Bold runs use a stroked text render mode rather than a second font file, which keeps the embedded subset small and the measurement table shared.

Justification is parsed from the stylesheet but rendered as left alignment in this build; the layout path that would distribute the extra space between words is a documented partial. The text below is long enough to page-break mid-paragraph so the boundary behavior is exercised by every run of the golden corpus.

2. Paragraph flow

Each paragraph starts on a new line and the margin between siblings collapses to the larger of the two. The content area is the page minus the configured margins, so a fixed A4 page at ten millimetres of margin gives the same column width for every document in this corpus.

When a paragraph is taller than the remaining space on a page, the engine moves the whole paragraph to the next page if it does not fit at all; if only the tail of the paragraph does not fit, the completed lines stay and the remainder flows onto the following page. Paragraph boxes that fit are kept whole on one page whenever the box layout allows it, which is what makes headings and their surrounding prose stay readable.

The result is that a report with two hundred paragraphs produces a continuous flow with no widows at the top of a page and no stray blank pages: the last line of a paragraph may end a page, but the first line of a paragraph never begins a page on its own unless the paragraph is shorter than one line.

3. Measurements at 96 dpi

All CSS lengths resolve against the 96 dpi reference grid: one pixel is three quarters of a point, millimetres convert at 25.4 millimetres per inch, and em lengths resolve against the current font size. The eleven point body text used here wraps to roughly ninety characters per line on an A4 page, which matches the classic typographic measure for print documents.

Readers of earlier drafts have asked whether the whitepaper should use a smaller measure to reduce the number of pages; the maintainers decided to keep the reference measure and let this document demonstrate the natural pagination of prose rather than optimize the page count.

The paragraphs that follow repeat the key claims so that the golden fixture stays comfortably multi-page even if the surrounding text is edited: a deliberate design decision for the corpus, where the fixture must never collapse to a single page. Each paragraph below is a full block of text with a subject sentence and supporting sentences, wrapped at the content width by the same line breaker that production documents use.

Line breaking happens at the whitespace between words, but the rule has subtleties that the test corpus pins down. A run of words with no spaces stays on one line until it exceeds the column width, at which point it overflows rather than being broken at an arbitrary character: there is no hyphenation pass in the engine, and the matrix documents that long unbroken tokens are a known limitation. Documents that contain long URLs or serial numbers therefore take care to insert break opportunities, typically by wrapping the token in a separate block or by shortening it before layout.

Letter-spacing is honored during measurement, so a paragraph that declares letter-spacing consumes more width per line and therefore wraps earlier than an identical paragraph without it. The engine measures the advance of every glyph at the resolved size and multiplies by the spacing factor declared in the stylesheet, which keeps the rendered line ends consistent with the measured ones; a mismatch here would produce text that overflows its box, and the corpus checks for that by asserting the page envelope of the typography fixture.

Font sizes resolve against the cascade: em and percent values refer to the parent font size, rem values refer to the sixteen pixel reference, and the keyword sizes (small, smaller, large, and so on) map to the standard scale. The line height defaults to the normal value, which the engine approximates as 1.2 times the font size, and can be overridden with a number, a length, or a percentage. Margins between block siblings collapse to the larger of the two, so a report with generous heading margins and modest paragraph margins gets the expected vertical rhythm without duplicated spacing.

Tables participate in the same flow: a table is a block-level box whose height is the sum of its rows, and a row never splits across a page boundary. When a table is taller than the remaining space, the whole table moves to the next page rather than leaving a fragment behind; this is the behaviour the fixture with the heavy invoice exercises, and it is why that fixture's page count is part of the corpus assertions. Cells with colspan consume the width of the columns they span, and cell content wraps at the cell's measured width, which is derived from the widest unbreakable content of any cell in the column.

Images are replaced elements: they occupy their intrinsic size scaled by any declared width or height, and they move wholly to the next page when they would cross a boundary. The image grid fixture embeds four data-URI PNGs at their intrinsic sizes and asserts that each produces a PDF image xobject; the logo fixture exercises the relative file path with local file access enabled, which is the configuration the corpus always uses. An image that fails to decode is skipped with a warning rather than aborting the conversion, and the broken-image behaviour is not painted.

Headers and footers are drawn into the margin band after the body pages exist, so the page-number placeholders in header and footer text are final when painted. The multi-section document fixture provides the multi-page body that header and footer flags draw on in the CLI smoke tests, and the header and footer text heights reserve the band during layout so body content never collides with the band. This is why the margin settings accept negative values to mean auto, measured from the header and footer heights.

The outline is built from the heading elements in document order, with the heading level preserved, and the outline depth flag truncates the tree at the requested level. The typography fixture exercises all six heading levels so the outline contains a six-level path; the table-of-contents objects consume the same tree to render the entry list with page numbers, dotted leaders, and forward and backward link annotations. The runner asserts that every fixture with headings produces a non-empty outline when the outline flag is on, which is the default.

4. Conclusion

In summary, pagination of long-form prose is line-level and box aware: rects split at page boundaries, while text lines, images and link annotations move whole. The behaviour is deterministic for a given input, which is exactly what a golden corpus needs: the page count and the structural assertions in the runner pin the behaviour across releases, and any change to the wrapping algorithm will show up as a page-count change in this fixture.

This final paragraph is intentionally wordy and repeats the document title - "How Long Documents Paginate" - a second time so that the searchable text of the PDF contains a stable marker string for any content-extraction check in future phases. The whitepaper is published under the project documentation umbrella and maintained together with the compatibility matrix that lists exactly which CSS properties are honoured by the layout engine.

One more paragraph: the corpus runner converts every fixture with local file access enabled, embeds the Liberation Sans subset, and asserts the PDF structure, the page envelope and the font object; any fixture that errors, produces zero pages, or violates the xref layout fails the build. That is the contract this whitepaper describes.

5. Appendix: measurement methodology

All measurements in this whitepaper were taken on an A4 page with ten millimetre margins on every side, the default geometry of the golden corpus. The content column is therefore always 190 millimetres wide - approximately 539 points - regardless of which fixture is being converted. The line breaker wraps text at that column, so the wrapping behaviour described in section one is reproducible on any machine that runs the corpus.

The corpus deliberately keeps every fixture free of JavaScript, of remote fonts, and of network dependencies: relative stylesheets and images resolve against the fixture directory with local file access enabled, and data URIs are decoded by the loader. This keeps the conversion deterministic and lets the golden runner run without network access, which is a requirement for CI environments that build the project behind a firewall.

Readers who want to extend the corpus with their own fixture should follow the established naming convention - fixture-NN-description.html - add the expected page envelope to the runner's bounds table, and verify the conversion with the command shown in the repository documentation. New fixtures should exercise at least one feature that the existing corpus does not cover, and they should state in their header comment exactly which behaviour they prove.

The final appendix paragraph repeats the key claim once more so the fixture stays safely beyond two pages even if a future refactor of the line breaker changes the wrapping density by a small margin: pagination of long-form prose is line-level and box aware, the output is deterministic for a given input, and every page of this whitepaper demonstrates the paragraph flow that production reports rely on. This is the last sentence of the appendix and of the whitepaper itself; it exists purely to give the fixture a stable length.